Unit 2: Using Informatics Principles, Systems, and Tools for Surveillance



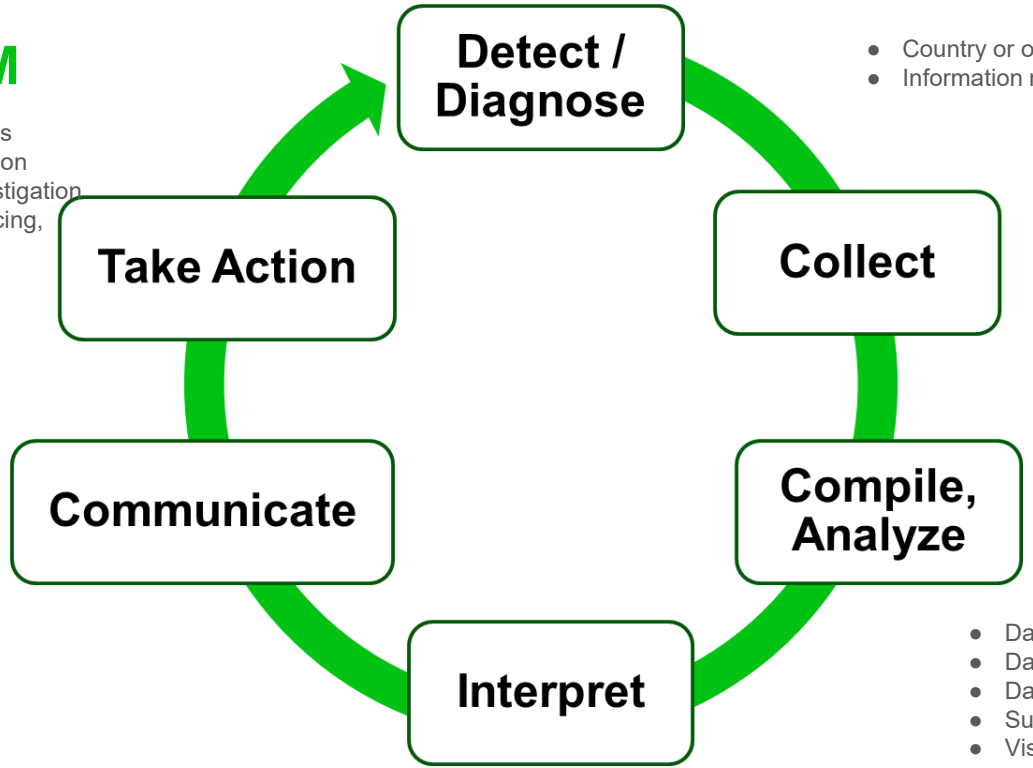
Laila's Surveillance Data & Analysis Problems

Laila must provide her supervisor with a **summary surveillance report** on the **current status of TB** in her district.

What gets in her way:

- Data collection and reporting can be unorganized, uncoordinated, and delayed.
- Usefulness of the data she has is limited because key variables and data are missing.
- Data from different sources is stored in multiple databases that must be merged.
- Unclear if the results are due to an actual increase in cases or other factors.

Creating Value from Public Health Surveillance Data



WISDOM

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

- Country or organizational context
- Information needs

- Data sources
- Data standards (variables, types, formats)
- Data quality
- Data security, privacy, & confidentiality

DATA

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

KNOWLEDGE

- Data organization & storage
- Data cleaning & transformation
- Data merging & exchange
- Summarize data
- Visualize data

- Describe and interpret data

INFORMATION

Unit 2: Goals

- **Re-use data** from existing information systems to detect, diagnose, and respond to outbreaks
- **Use common software features and functions** to explore, describe, and prepare data for analysis
- **Identify and use software features and functions** to detect and address data quality issues
- **Apply informatics standards and software features** to organize and analyze surveillance data
- **Use basic data visualization and reporting features** to prepare a surveillance summary report.

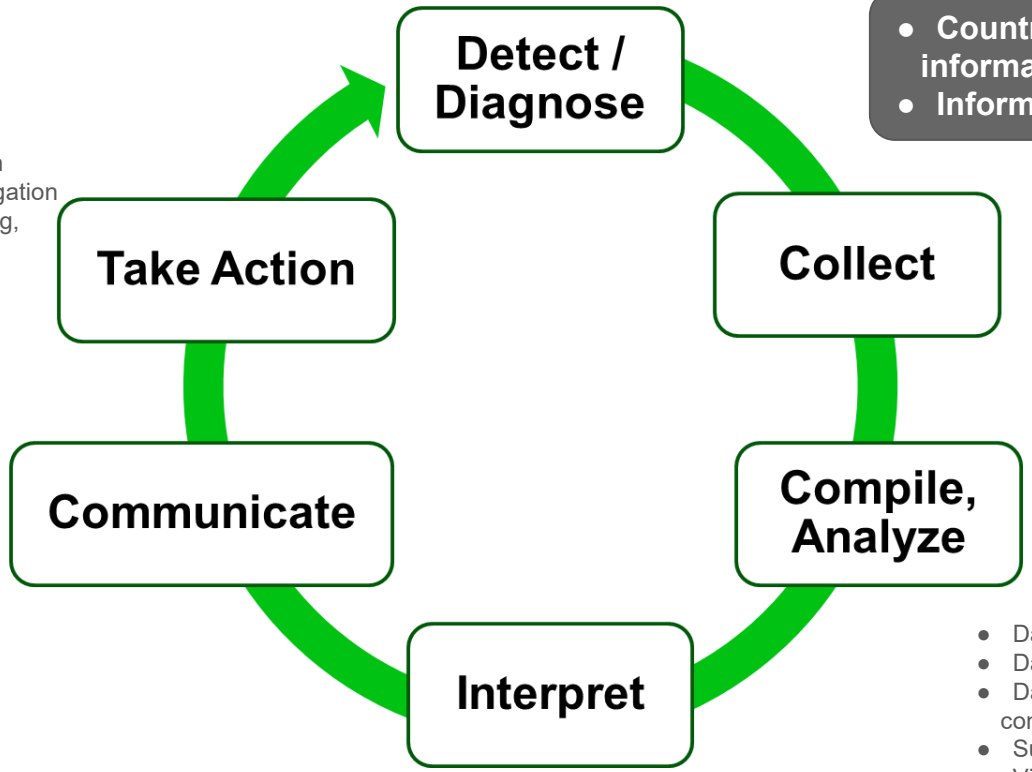
Unit 2 Case Study Assignment

Your report will need to answer the following key questions:

- How many cases were detected? Has the epidemic threshold for measles been crossed?
- Where did the cases occur?
- Who is affected?
- Is the increase due to an outbreak? Is there another explanation for the increase in reported cases?

Unit 2.1:
Using Informatics Principles, Systems, and Tools to Access, Explore, and Prepare Your Data for Analysis

Creating Value from Public Health Surveillance Data



- Country, organizational, health information system context
- Information needs

- Data sources
- Data standards (variables, types, formats)
- Data quality
- Data cleaning & transformation

- Data organization & storage
- Data merging & exchange
- Data security, privacy, & confidentiality
- Summarize data
- Visualize data

- Describe and interpret data

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

Where Laila is now



Where Laila wants to be

Current State

Data collection and reporting can be unorganized, uncoordinated, and delayed.

As a result, Laila has limited access to databases and finds herself working very large datasets.

These datasets do not always include the data she needs.

In order to meet her information needs, she needs to locate and access additional data in other systems and datasets.

Future State

Rapid, easy access to a dataset of quality data that responds to Laila's specific information needs.

Unit 2.1: Learning Objectives

- Define the information needs for public health surveillance
- Discuss how informatics principles and systems address common challenges with collecting surveillance data
- Identify and evaluate existing country information systems to determine if and how data can be re-used for public health surveillance.
- Securely access selected databases and obtain data required for public health surveillance
- Use a standard data dictionary and software features to explore surveillance data

Where Laila is now



Data collection and reporting can be unorganized, uncoordinated, and delayed.

As a result, Laila has limited access to databases and finds herself working very large datasets.

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What causes this situation?



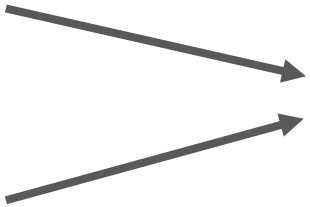
Two root causes that informatics help solve

1. Multiple, uncoordinated health information systems and software used to collect public health data for different stakeholders that have overlapping information needs
2. Technology infrastructure issues (ie: connectivity, power outages, etc) interfere with data collection and reporting

Using Informatics to Make Decisions and Take Action

Information needs for surveillance

Country health information system context



What existing software applications are used to collect and store the data needed for analysis

COMMUNITY HEALTH TOOLKIT

dhis2 Tracker

ei

sormas

CommCare

ODK

REDCap Research Electronic Data Capture

KoboToolbox

OPENELIS GLOBAL

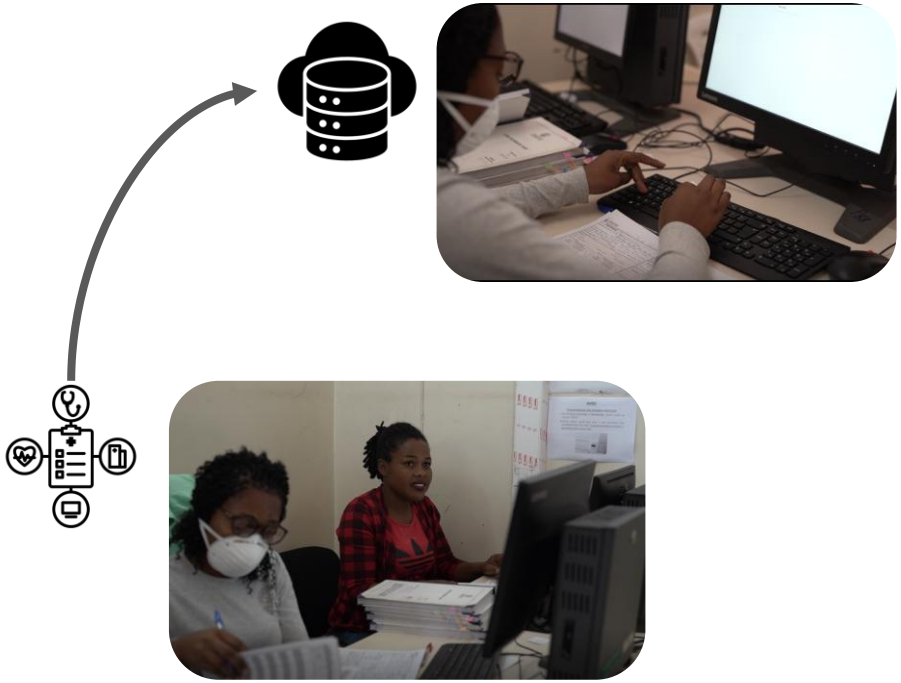
OpenMRS MEDICAL RECORD SYSTEM

Using Informatics to Make Decisions and Take Action

Support health care workers at the community and facility level….

….to use **existing health information system** software….

….to **collect and report** complete surveillance data on time.



Understanding the Country & Health Information System Context



Understand the Health Information System Context

Understanding the *HIS* context can determine if and how to engage with the **software applications and innovations** used within a country's **health information system**.

The health information system context is

- fluid, multi-faceted, and ever-changing;
- defined by the political and economic environment, technology infrastructure, and other factors;

Root Cause #1: Multiple information systems are used to collect and manage public health data.

Multiple paper and electronic systems have been used to record and report data for clinical, epidemiological, research, and administrative purposes by different programs and stakeholders.

This leads to:

- High data collection and reporting burden
- Duplication of data collection efforts
- Low or no perceived value to data collection
- Poor data quality
- Low data use

Understanding the HIS Context

- What is happening in the political and economic environment that influences the HIS software applications used at all levels of the system?
- What software applications are currently being used at each level of the system?
- Which HIS software applications manage the data needed for surveillance and outbreak investigations?
- Who uses these HIS software applications?

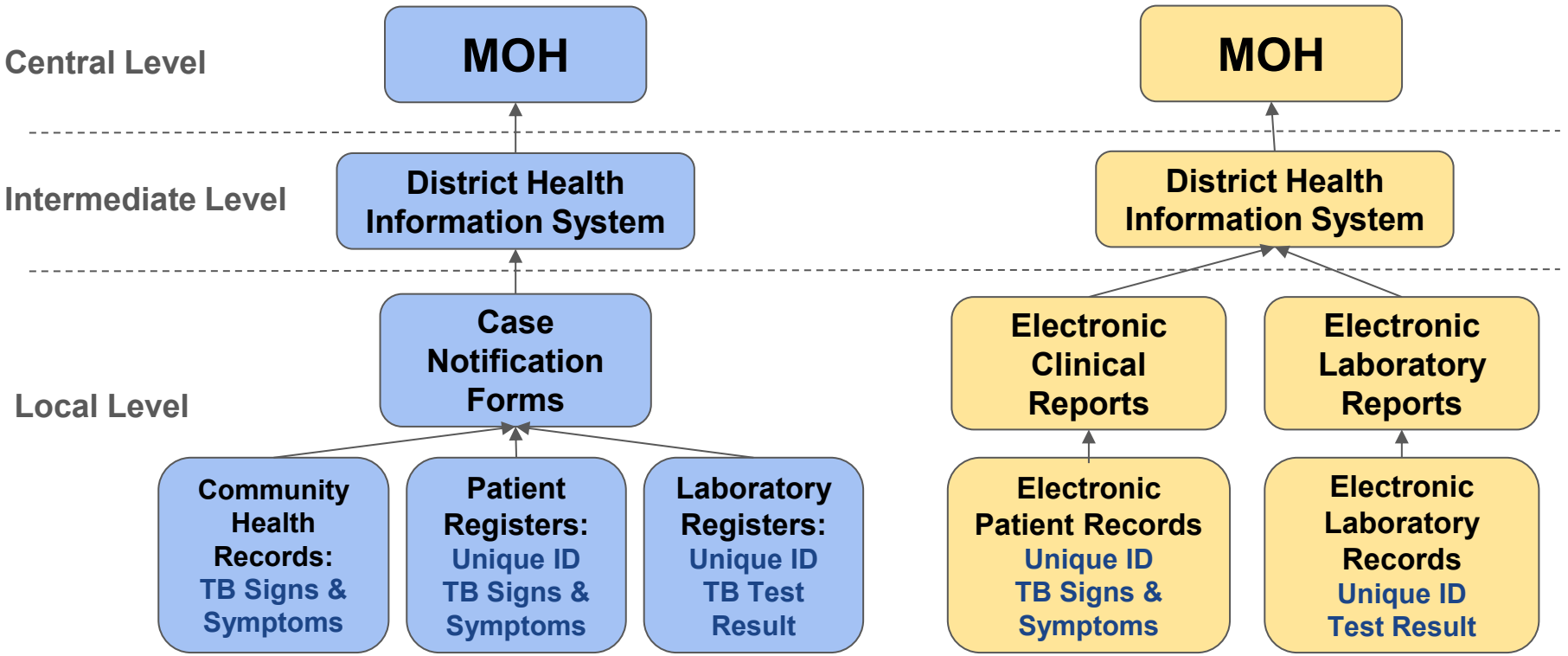
Key Changes in the HIS Context

Countries are

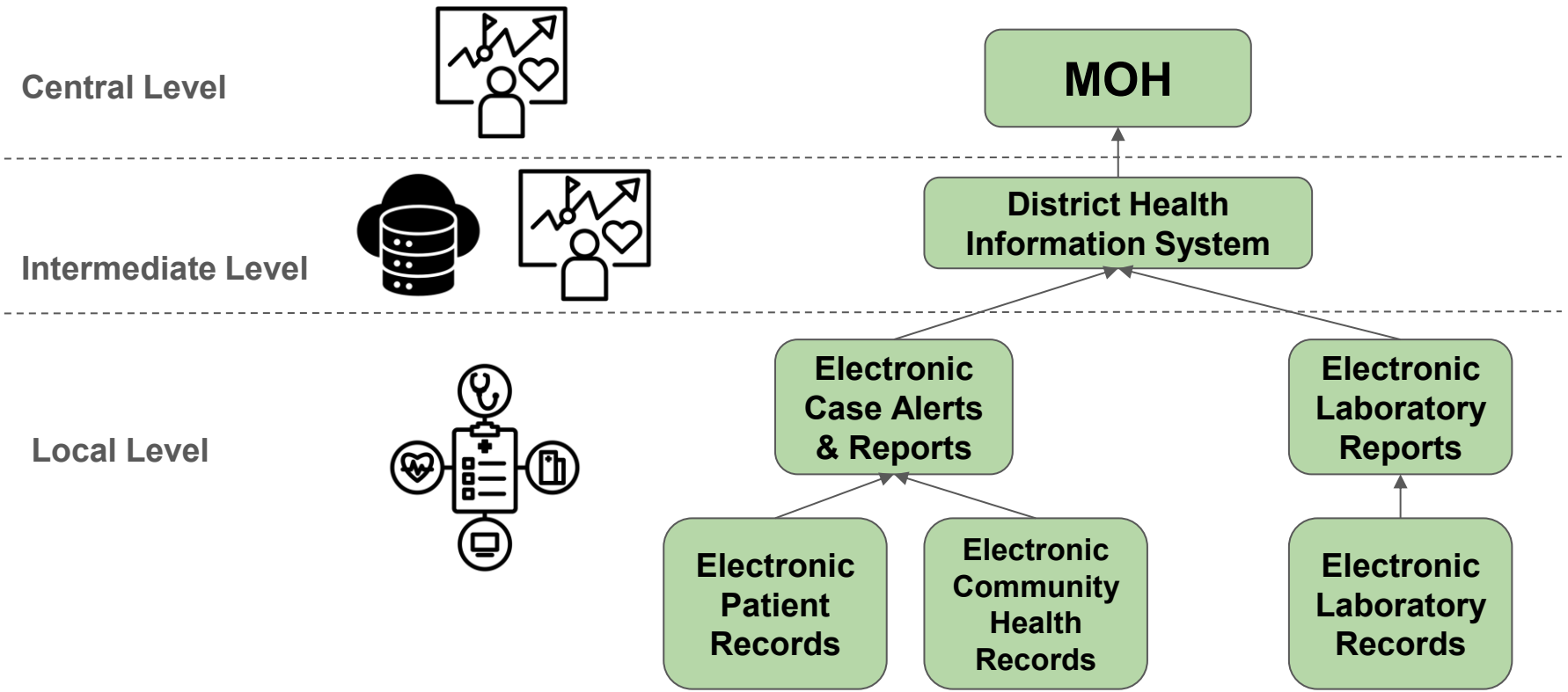
- moving towards streamlined, electronic information systems for public health and surveillance.
- taking greater ownership of health information systems, including surveillance systems, and the data they collect.
- implementing national policies and strategies that increasingly influence the software applications used to manage public health data.

Electronic Disease Surveillance System Systems

Health Information



Technology + Country IDSR + Health Information Systems





Laila's Country HIS Context

Central Level

District Level

Health Facility & Community Level

MOH

dhis2

District Health Information System

Electronic Case Alerts & Reports

dhis2

Electronic Laboratory Reports

OpenMRS MEDICAL RECORD SYSTEM

Patient Records

Community Health Records

COMMUNITY HEALTH TOOLKIT

Laboratory Records

OPENELIS GLOBAL

By moving towards standard, electronic information systems for public health *and* surveillance, countries aim to :

- Enable better data transmission and management, including data storage and easy access
- Reduce data collection and system costs
- Easily generate automated alerts for epidemiological and clinical action
- Assist in early detection, investigation, and response to outbreak or public health events
- Enhance virtual, near real-time disease monitoring capability

Small Group Work

In your small group,

1. List the information Laila will need for her TB surveillance summary report.

1. Using Laila's country HIS diagram, identify:
    - Where the data Laila needs originates
    - What paper and electronic systems are used to collect and store this data
    - Who has access to and uses the paper and electronic systems

Be ready to share your work in 10 minutes



Laila's Problem: Getting the data she needs

Laila knows

- the information she needs for her TB surveillance report, and
- what system applications to use to retrieve this data from the right databases.

As Laila goes to log on to the system, she discovers that the system is offline.

What might cause the system to be offline?

Root Cause #2: Issues with Technology Infrastructure

Poor connectivity, power outages, and software application issues (ie: crashing, downtime) interfere with

- data collection,
- electronic reporting,
- compilation, and
- analysis and visualization

How can understanding a country's HIS context address these challenges?

Information Technology Support Systems and Policies

- IT support or “Help Desk” systems with a workforce who can respond to user issues
- Data management procedures that include guidance on:
  - Backup and recovery
  - Hardware to manage power instability (surge protectors, generators, etc)
  - Emergency response plans
  - Tools to monitor system performance



Actions Laila can take

What actions can Laila take in order to get the data she needs while the system is offline?

- Follow the Help Desk procedures
  - Contact the Help Desk to report the issue
  - Get troubleshooting assistance
- Obtain a copy of the backup database

Using Informatics to Make Decisions and Take Action

Once Laila understands her country's broader HIS context, she can apply her knowledge to:

- Identify and use the HIS software that meets her needs
- Follow up with and support healthcare workers to effectively use existing software applications to collect surveillance data, report it on time, and improve the quality of the data
- Use country IT procedures and policies to navigate technology infrastructure challenges

Case Study: Surveillance

Information Needs
Understanding the HIS context

- Define the information needs for public health surveillance
- Determine what existing country information systems will meet your surveillance information needs
- Use informatics concepts, systems, and tools to collect or generate the data for your surveillance summary report

Case Study Context

On 25 October 2021, your supervisor received a call from the Chief Medical Officer at the Kukuna Health Center, informing her of a growing number of recent reports of fever and rashes coming from community health care workers within the area. One of your primary responsibilities is to detect outbreaks of fever with rash for investigation.

Your supervisor asks you and your team to prepare a summary surveillance report on the current status of measles.

Context: Sierra Leone IDSR Strategy

Approach: Improve public health surveillance and response for priority diseases, conditions and events at community, health facility, district and national levels through the **integration** and **coordination** of common surveillance activities and functions.

Integration includes the harmonization of different methods, software, data collection forms, standards and case definitions in order to prevent inconsistent information and maximize efforts among all disease prevention and control programmes and stakeholders.

Coordination involves information sharing, joint planning, monitoring and evaluation to provide accurate, consistent and relevant data and information to policy-makers and stakeholders at district and national levels.

The district level is the hub and focus for integrating surveillance functions.

District Activities and Responsibilities

You recently arrived in Sierra Leone and have been assigned to one of three District Health Management Teams.

At the district level, you and your team's responsibilities include:

- Collecting and aggregating surveillance data from health facilities and the community
- Reviewing the quality of surveillance data
- Analyzing data by time, place, and person
- Updating graphs, tables, and charts to describe reported diseases, conditions, and events on a weekly basis
- Comparing data and make conclusions about trends and thresholds
- Reporting data on time to the province/regional surveillance officer

Sierra Leone: Priority diseases, conditions, and events

Epidemic-prone diseases, conditions or events which require immediate reporting	Diseases targeted for eradication or Elimination	Other major diseases, events or conditions of public health importance
1. Acute haemorrhagic fever syndrome*	1. Buruli ulcer	1. Adverse drug reaction
2. Acute jaundice syndrome	2. Dracunculiasis (Guinea Worm)	2. Diabetes mellitus (new cases)
3. Adverse events following immunization (AEFI)	3. Leprosy	3. Diarrhoea with dehydration less than 5 years of age
4. Animal Bites (Dog and Snake bite)	4. Lymphatic filariasis	4. Epilepsy
5. Anthrax	5. Measles	5. HIV/AIDS (new cases)
6. Bacterial Meningitis	6. Neonatal tetanus	6. Hypertension (new cases)
7. Chikungunya	7. Poliomyelitis***	7. Injuries (road traffic accidents)
8. Cholera	8. Onchocerciasis	8. Malaria
9. Dengue fever		9. Non-neonatal tetanus
10. Diarrhoea with blood (Shigella)	**Diseases or events of International concern**	10. Perinatal deaths
11. Human Rabies	1. Human influenza due to a new subtype***	11. Severe Malnutrition in children under 5 years of age
12. Maternal deaths	2. SARS***	12. Severe pneumonia less than 5 years of age
13. Monkey pox	3. Smallpox***	13. STIs
14. Plague		14. Schistosomiasis
15. SARIs**	Any public health event of international or national concern (infectious, zoonotic, food borne, chemical, radio nuclear, or due to unknown condition.	15. Soil transmitted helminths
16. Typhoid fever		16. Tuberculosis (new cases)
17. Yellow fever		17. MDR/XDR Tuberculosis
18. Zika virus disease		

Case Study: Informatics and Surveillance System Context

The application of informatics, specifically electronic tools, is considered a platform for enhancing real-time surveillance, facilitating prevention, prediction, detection, reporting, and response.

In Sierra Leone, the application of informatics to the existing surveillance system is based on:

- standardized interoperable and interconnected information systems used within the national context
- rapid collection, analysis, reporting, and use of disease/events data in real-time for the right public health action.

Resources: Sierra Leone eIDSR and HIS Strategies

The following resources describe the current eIDSR and HIS context, strategies, and policies:

- Technical Guidelines for IDSR, Section 11
- Situational analysis of HIS in Sierra Leone, Sierra Leone Health Information System Policy, December 2021
- Community Health Information System (CHIS), National CHW Policy 2021

Unit 2 Case Study: Surveillance

Case Study Assignment #1

With your small group, use the next 15 minutes to:

- Review the case study introduction
- Answer Case Study Part 1 Questions 1 and 2

Be prepared to report back on your findings.

Using Software to Explore Datasets



Laila's Problem: Getting the data she needs

Laila's supervisor gave her an Excel file with a dataset of over 8,500 rows of surveillance data.

It includes 35 different variables.

Laila does not know if this dataset includes the variables and data she needs.



What might cause Laila’s problem?

The Excel backup includes all surveillance data for the country for the past five years.

Use epidemiology and software to get the right data

To make data more useful, ask the following questions:

- What priority conditions does her dataset include?
- What timeframe does her dataset span?
- What locations does her dataset include?
- Who is included in her dataset?
- Does the dataset include the priority condition, timeframe, locations, and population that she is most interested in?

How Excel is used by epidemiologists

Excel databases are organized similar to a line list:

- A structured set of data
- Organized in table with headers on the top row and related data or values in each column.

Easy to use as a database for simple explore, store, manage, and analyze surveillance data



Five Essential Excel Functions

- Sort: organizes one or more columns of data in ascending or descending order
- Filter: shows or hides specific data in a list
- Count: displays the number of cells in a range containing numbers
- Max: displays the largest value in the cell range
- Min: displays the smallest value in the cell range

Excel How-To Scribes

What are Excel How-To Scribes?

Easy to follow, step by step instructions with screenshots

What Excel How-To Scribes are available?

1. Make a Working Copy of Your Dataset
2. Create a Copy of a Dataset in a New Worksheet
3. Sort Data Using a Single Variable
4. Sort Data Using Multiple Variables
5. Filter Data
6. Use the Count Function in Excel
7. Use the Max-Min Function

Unit 2 Case Study: Surveillance

Case Study Assignment #2

Use the next 15 minutes to:

- Use Excel and the Excel How-To Scribes to explore your dataset
- Answer Question #3

Be prepared to show how you arrived at your responses.

Standards and Data Dictionaries



Laila's Problem: Getting the data she needs

Laila used Excel to create a smaller dataset for the condition, time period, and location she is most interested in.

It is unclear if her dataset includes the right variables she needs for her report.

- Do the variables have the same meaning?
- Are they the type and format she needs?



What might cause Laila’s problem?

- Different countries and systems use different data dictionaries
- Data dictionaries are incomplete or are not consistently used

Informatics Solution: Applying Standards

Using standards in public health provides key benefits:

- Standards ensure that data is collected and reported in a uniform way across different systems, regions, and time periods.
- Standardized data definitions and formats help reduce errors and misinterpretations, improving the reliability of the data used in public health decision-making.
- Standards promote consistency and reduce errors, minimizing data quality issues and leading to fewer inconsistencies in case definitions, disease classifications, or health metrics.

In Practice: Decisions and Action

Epidemiologists can apply their knowledge of the country health information system context and data standards to:

- Identify and consistently use standard data dictionaries to select, organize and prepare the data they need for surveillance and outbreak investigation
- Help make data systems and datasets more responsive to their information needs by suggesting improvements to existing data dictionaries
- Identify and engage with any groups that maintain standard data dictionaries

Scenario

The dataset Laila is working with has a variety of data quality issues. These issues stem from both individual records and HIS system software.

Some data is missing, incorrect, or illegible. There may also be duplicate records or missing data due to delayed data entry, reporting, or simply the way the data system is designed.

Scenario

If Laila uses the same data dictionary as other epidemiologists and informaticians throughout the country, then

- Epidemiologists like Laila can trust that the surveillance data collected throughout the country has the same meaning, type, format, and response options.
- Informaticians can set up HIS systems to consistently collect essential data that Laila needs for routine surveillance

Why Use a **Standard** Data Dictionary?

When health information system software use a shared, standard data dictionary,

- variables used across systems and the country share the same meaning, name, type, and format
- uniformity in data representation reduces errors, improving data quality
- data collected by different sources can be exchanged by different health information system software

What are data standards?

- Rules and guidelines set to **ensure consistency** and **reduce errors** during data collection.
- Data standards are often included as components of data dictionaries:
  - Unique identifiers
  - Standard codes, such as ICD-10, Loinc
  - Master facility lists

Why Use a **Standard** Data Dictionary?

Data dictionaries provide a common language and understanding of:

- The meaning of collected data.
- Variable names and types.
- Response options and codes (yes/no).
- How the data was collected

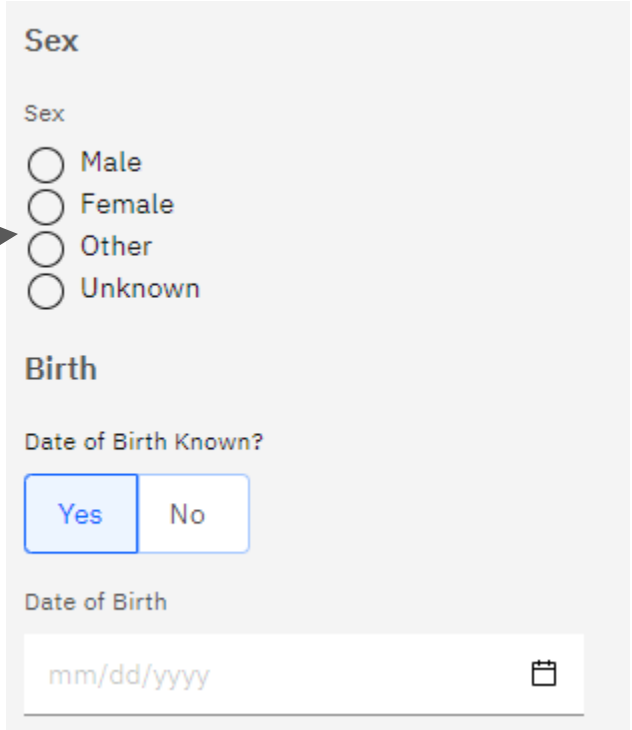
Key Data Dictionary Components

Variable Name	Field Type	Data Element Description	Response Options, Allowable Values/Value Set
patient_ID	string	The unique national health service identifier for this patient.	
first_name	string	The given name of the patient.	
last_name	string	The surname of the patient.	
date_birth	string	The date of birth of the patient in YYYYMMDD format.	
sex	char	Patient sex at birth.	{‘M’, ‘F’, ‘O’, ‘U’}
race	char	Patient’s race category in compliance with US OMB census rules.	LOINC Race or Ethnicity 46463-6 value set
notifiable_diseases	list[string]	A list of notifiable diseases that is currently relevant to disease surveillance.	WHO and National IDSR priority diseases, conditions, and events value set

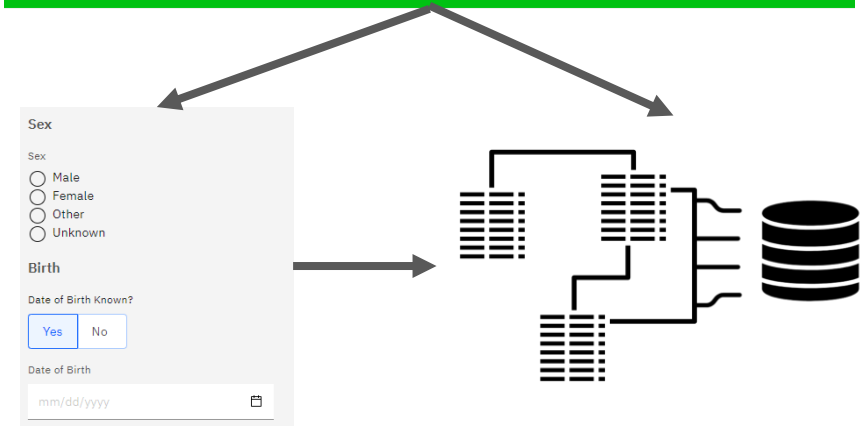
How HIS System Software Uses Data Dictionaries

Informaticians use the data dictionary to program data entry fields with the correct variable name, field type, and response options

Variable Name	Field Type	Data Element Description	Response Options, Allowable Values/Value Set
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sex	char	Patient sex at birth.	{'M', 'F', 'O', 'U'}



Data dictionaries also describe how the data are arranged in computer files or databases.

Unit 2 Case Study: Surveillance

Case Study Assignment #3

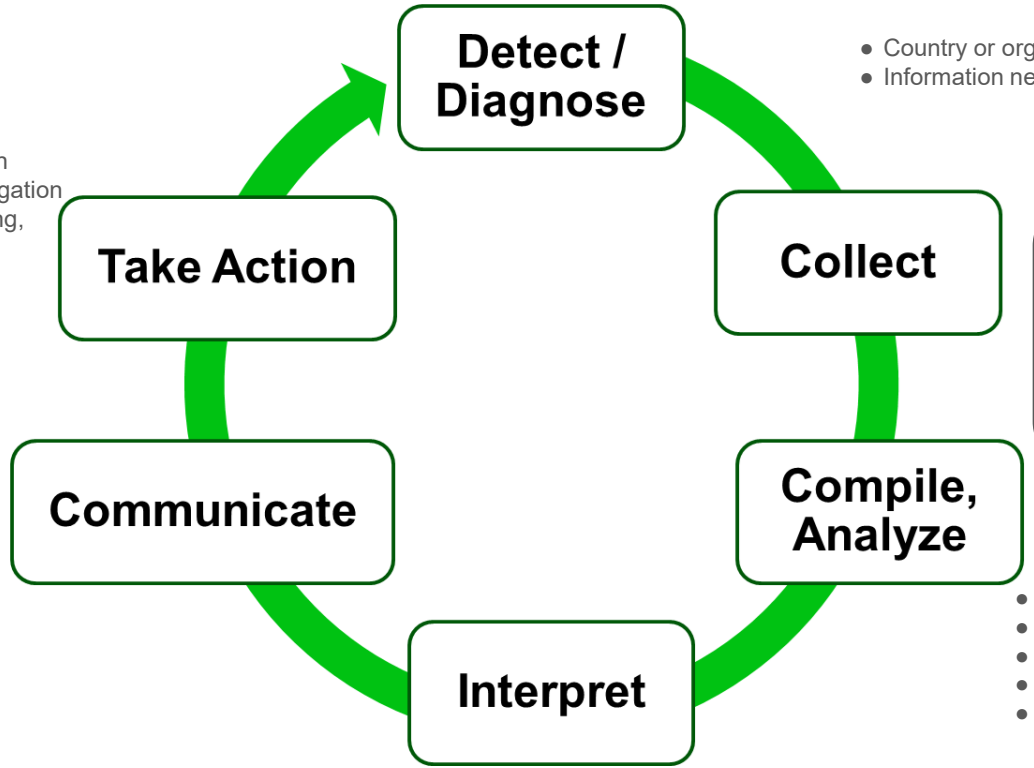
In your small groups:

- Use Excel and the data dictionary to further explore your dataset
- Answer Questions 4 and 5

Be prepared to share your responses.

Unit 2.2:
Using Informatics Principles and Tools to Detect and Fix Data Quality Issues

Creating Value from Public Health Surveillance Data



- Country or organizational context
- Information needs

- Data sources
- Data standards (variables, types & formats)

➔ **Data quality**

➔ **Data cleaning & transformation**

- Data organization & storage
- Data merging & exchange
- Data security
- Summarize data
- Visualize data

- Describe and interpret data

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

INFORMATION

Where Laila is now



Data collection and reporting can be unorganized, uncoordinated, and delayed.

- As a result, Laila has limited access to databases and finds herself working with very large datasets.
- These datasets do not always include the **quality** data she needs.
- In order to meet her information needs, she may need to locate and access additional data to address **missing** data.

Dataset now includes the right variables.

Is the data the right type and in the right format?

Is the data in the cells complete, accurate, and timely?

Is there any missing data she needs to follow up on?

What causes poor quality data?

Data entry

- Missing data
  - delayed data entry
  - skipped fields
- Incorrect data
  - illegibility
  - typos
  - transcription errors
  - transposition errors

HIS system software

- Delayed records
- Missing data values and types
- Duplicate records

Informatics Solutions Targeting Data Quality

- Healthcare workers across the country consistently use the same HIS software to collect and report surveillance data
- Data dictionaries **standardize data types and formats**
- Software applications include
  - **required fields and response options** based on data standards
  - **data quality checks** or validation rules

Scenario: Martha's Experience Three Years Ago

Martha is a Community Health Care Worker assigned to ten communities in her catchment area. In addition to surveillance data, Martha collects and reports data for multiple partners representing different programs: HIV, Maternal/Child Health, TB, Malaria, NCDs.

Three years ago, Martha had to collect and report data using a different application on different tablets provided by partners supporting each of these programs. As a result, she spent most of her time doing double or triple data entry and reporting.

How does this impact the quality of the data used by epidemiologists at the district level?

Scenario: Martha’s Experience Today

Since the MOH began implementing their 2022 HIS Strategy, Martha received training on and started using a new, standard Community Health Care app. This app replaced several of the other software applications she used to use.

Scenario: Martha's Experience Today

The standard community health information application has several features that support data quality:

- Requires completion of a core set of data fields, such as diagnosis, onset date, residence, and others.
- Uses drop down lists for fields like notifiable conditions and pick calendars for dates
- Shows a pop up alert when she enters data that is outside the expected range

How will Martha's use of this new application impact epidemiologists like Laila?

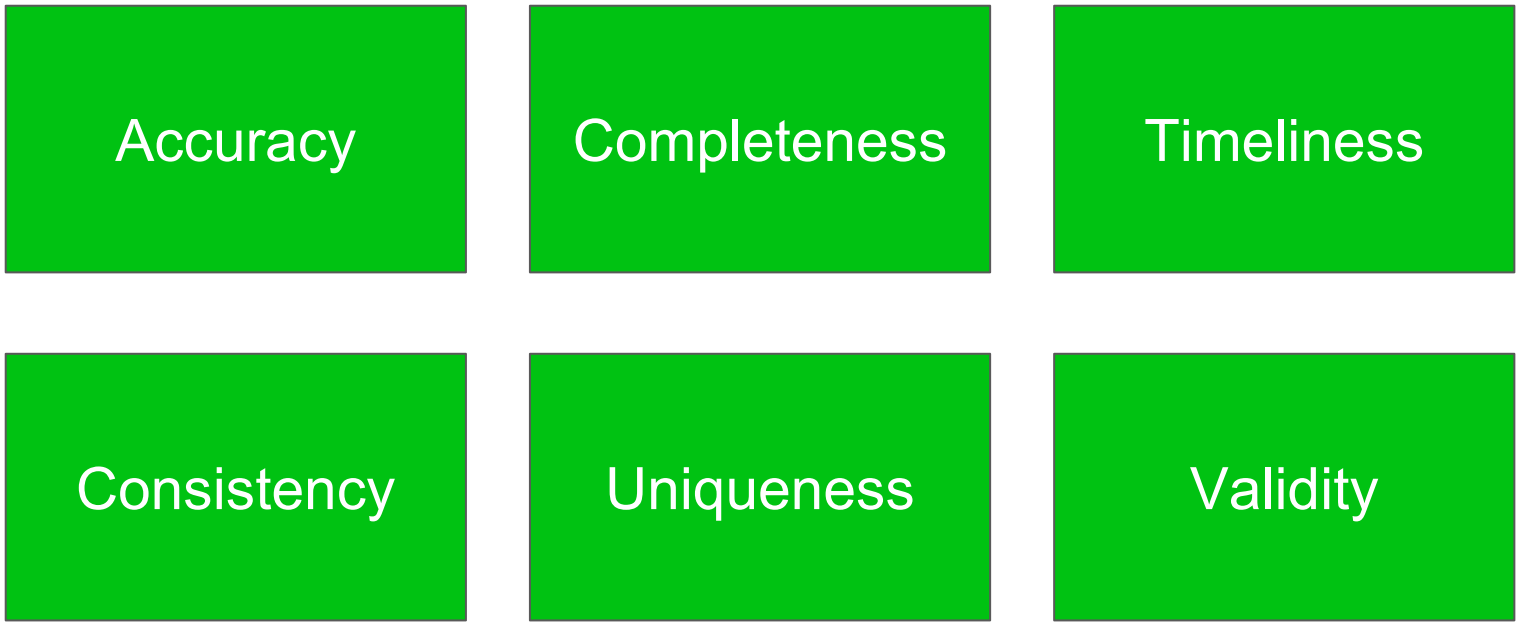
What is data quality?

Definition

The level of accuracy and completeness of data in a dataset

- Do the data accurately reflect reality, so that the data serve the intended purpose?
- High quality data more likely to lead to better decision-making and planning

Dimensions of Data Quality



Dimensions of Data Quality: Accuracy

Accuracy

Definition
The extent to which data correctly describes the real-world object or event it represents. Accurate data is free from errors and correctly reflects the truth.

Data comes from sources that are verifiable and trustworthy.

Informatics Solution
HIS systems use a data dictionary to standardize variables, reducing data entry errors.

Dimensions of Data Quality: Completeness

Completeness

Definition
The degree to which all required data is present.

Complete data contains all necessary elements and values needed for its intended use.

Informatics Solution
HIS software that use standard data dictionaries are able to capture all necessary data, including metadata.

This may include the use of required fields.

Dimensions of Data Quality: Timeliness

Timeliness

Definition
The degree to which data is up-to-date and available within a suitable time frame.

Timely data ensures that information is current and readily accessible when needed.

Informatics Solution
HIS software can

- document the date and time of data entry
- display alerts when specific conditions are met
- automatically generate and transmit reports

Dimensions of Data Quality: Consistency

Consistency

Definition
The degree to which data is uniformly consistent within a dataset and across different datasets - and systems.

Consistent data does not have conflicting or redundant values, allowing integration and comparison of data from multiple systems.

Informatics Solution

- Data dictionary
- Health care workers are trained to use HIS systems for data collection and reporting **in the same way.**

Dimensions of Data Quality: Uniqueness

Uniqueness

Definition

The extent to which each record is unique within a dataset.

Unique data ensures that there are no duplicate records and each entry is distinct.

Informatics Solution

- Unique identifiers to create distinct, individual records and integrate unique patient records across HIS systems.
- HIS software features that can be used to detect and fix potential duplicate records

Dimensions of Data Quality: Validity

Validity

Definition

The degree to which data conforms to the defined formats, rules, and constraints.

Valid data adheres to predefined standards and business rules.

Informatics Solution

- HIS software captures data as defined in standard data dictionaries
- Data fields are designed to use standard formats and response options
- HIS software uses standard reporting formats

The role of informatics in data quality



Applying informatics to improve data quality

How you and health care workers can use software applications and policies or protocols to:

- Transform data => standardize data formats
- Clean data => deduplicate records, address missing data and data entry errors

How can you identify data quality problems using Excel?

- Use graphs to spot outliers (high/low values)
- Pivot tables to create summary tables of large datasets
- Filter and sort dataset to spot outliers
- Conditional formatting to spot outliers or invalid data

Data Quality Problem: Variable Coding

- Dataset includes a categorical response coded Yes/No
- Data dictionary uses a standard numeric response coded Yes = 1, No = 0

Solution: Using Excel to Transform Data

Solution:

- Recode variable to be consistent with the data dictionary
- Use Excel to convert categorical responses into numeric codes

Steps:

- Add a new column for the recoded values
- Use Excel’s IF function to assign numeric codes to categorical data
- Formula: =IF([Response=”Yes”, 1, 0)

Data Quality Problem: Duplicate Records

You want to make sure that there are no duplicate records in your dataset.

Solution: Use Excel’s “Remove Duplicates” feature

Steps:

- Select your data by highlighting the range of data that you want to check for duplicates.
- Go to the Data tab on the Excel ribbon
- Click on Remove Duplicates in the Data Tools group
- Configure Duplicate Removal. A dialog box will appear with all the columns selected by default. Unselect the columns that you do not want to check for duplicates.
- Review results. Excel will show a summary of how many duplicate records were found and removed.

Data Quality Problem: Relationship between fields is not logical

Solution: Check if the Date of Discharge (C2) is later than Date of Admission (B2)

Steps:

- Use a formula in a new column OR use conditional formatting
  Formula: =IF(B2>C2, TRUE, FALSE)
- Highlight or flag rows where the relationship doesn’t hold

Case Study: Surveillance

Unit 2.3: Individual Assignment #4

Use the next 30 minutes to respond to Questions 6 & 7 using Excel and your data dictionary:

- Review your dataset and identify data quality problems
- Outline the steps to take to fix each problem
- Fix identified data quality issues

Be prepared to share your work

Unit 2.3:
Compile and Analyze Surveillance Data

Where Laila was



Where Laila is now

Past State

Data collection and reporting can be unorganized, uncoordinated, and delayed.

As a result, Laila has limited access to databases and finds herself working very large datasets.

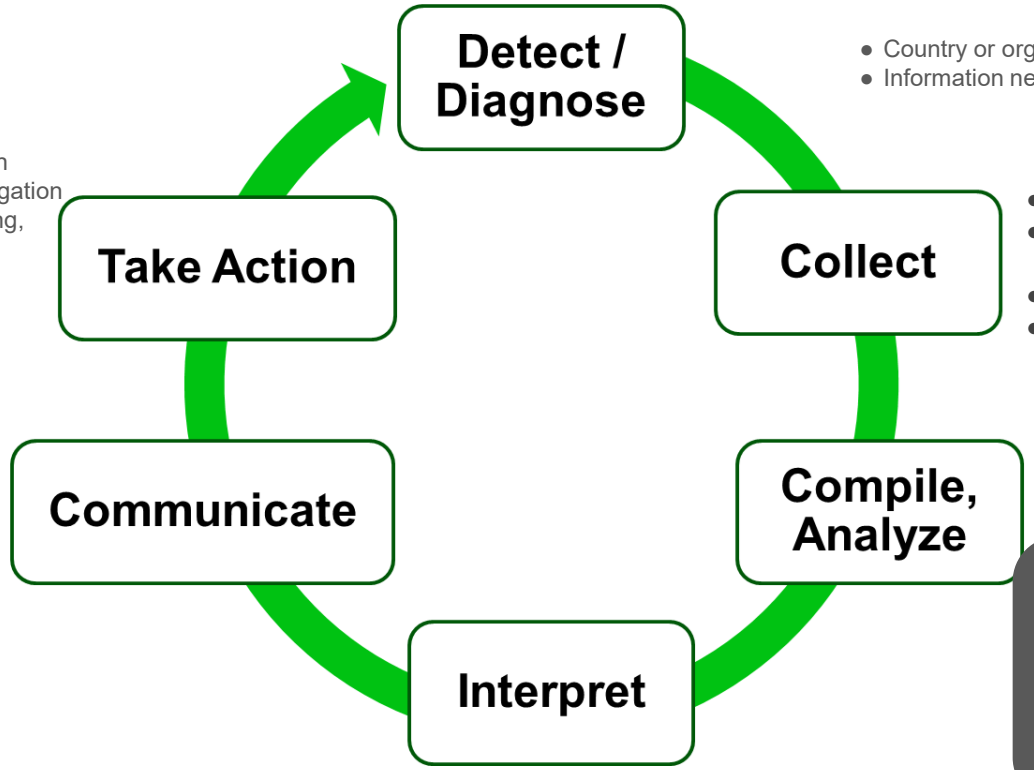
These datasets do not always include the data she needs.

In order to meet her information needs, she needs to locate and access additional data in other systems and datasets.

Current State

Access to a dataset of quality data that responds to Laila's specific information needs.

Creating Value from Public Health Surveillance Data



- Country or organizational context
- Information needs

- Data sources
- Data standards (variables, types & formats)
- Data quality
- Data cleaning & transformation

- Data organization & storage
- **Data merging & exchange**
- **Summarize data**
- **Visualize data**
- Data security

- Describe and interpret data

- Disseminate summary of surveillance data
- Alerts about outbreaks or events

- Impact, situation awareness
- Decision on short-term action
- Information needs for investigation
- Resource allocation (financing, workforce) for investigation

Where Laila is now



Data collection and reporting can be unorganized, uncoordinated, and delayed.

- As a result, Laila has limited access to databases and finds herself working very large datasets.
- These datasets do not always include the **quality** data she needs.
- In order to meet her information needs, she may need to locate and access additional data to address **missing** data.

Her current dataset includes the right case notification data

Data quality issues have been identified and fixed.

There are still missing variables and data to follow up on.



Laila’s Information Needs

- Person
  - unique identifier
  - age
  - sex
  - outcome/status at time of report
  - signs and symptoms
  - laboratory result
- Place
  - health facility
  - district
  - district of residence
- Time
  - date HF notified
  - date of onset

What data does Laila have?

What data does she still need?

What HIS systems generate this data? Where is it stored?



Laila's Problem and Its Causes

Problem

Different types of data from different sources are stored in multiple databases that must be merged.

Causes

- Multiple, uncoordinated HIS systems are used to process public health data
- Some HIS systems use different identifiers.



Decisions and Actions Laila Can Take

- Use datasets and data dictionaries from existing, shared databases at the district and national levels
- Use unique identifiers and other demographic variables to match cases across datasets
- Identify what software applications and their features and applications to use to merge one or more datasets
- Use software features to merge datasets or upload additional data into a district/national database



Laila Takes Action

After following up with the district data manager, she now has a dataset with yellow fever clinical data from the health facilities in the district. She is still waiting for data from the laboratory.

Laila uses Excel to sort the clinical dataset by unique patient identifier. Her notification dataset includes a unique case identifier. Both datasets include the patient age, sex, and date of onset.

Laila can use Excel to organize the two datasets and merge them together. From there, Laila can use Excel to review her merged dataset and resolve any new data quality issues.

Why merge datasets?

- Merging datasets combined two or more different datasets, creating a “wider” combined dataset.

- Each data source may store its data in its own database. To conduct your analysis, data from these sources must be stored in a single database.

Actions to take before merging datasets

- Check data quality and address any issues.
- Check that each dataset is structured, formatted, and organized the same way.
- Identify which fields are common to both datasets (Case ID, Date seen at HF/Community, Age, Sex, Date of Onset)
- Check that the values in both datasets align.
- Check if you need to add columns (widening your dataset) or add rows (lengthening your dataset)

When to widen a dataset

- A wide format has one case per row and one variable per column
- Widen your dataset when you want to add rows or variables
- Most useful for data exploration and display purposes

When to lengthen a dataset

- A long format has one row per time period (day/week/month)
- Lengthen your dataset when you need to add one or more variables to each case
- Most useful for processing, analysis, modeling, and data transformation

Example: Yellow Fever Datasets

Yellow fever suspected case definition: Any person with **acute onset of fever**, with **jaundice appearing within 14 days of onset** of the first symptoms.

Yellow Fever Case Notification Data

Case ID	Health Facility	Date seen at HF/Community	Age	Sex	Date of Onset
NMC002815	Kambia Government Hospital	2023-08-16	16	MALE	2023-08-11
NMC048650	Connaught Hospital	2023-11-13	55	MALE	2023-11-04
NMC059540	Kambia Government Hospital	2023-12-28	16	MALE	2023-12-20
NMC065919	Mayakie MCHP	2023-07-27	6	FEMALE	2023-07-20
NMC092898	Foredugu MCHP	2023-11-15	4	FEMALE	2023-11-15
NMC132107	Rokupr CHC	2023-08-06	40	MALE	2023-07-28
NMC144004	Connaught Hospital	2023-09-10	25	MALE	

Adding Demographic and Clinical Data

Case Notification Data

Case ID	Health Facility	Date seen at HF/Community	Age	Sex	Date of Onset
NMC002815	Kambia Government Hospital	2023-08-16	16	MALE	2023-08-11
NMC048650	Connaught Hospital	2023-11-13	55	MALE	2023-11-04
NMC059540	Kambia Government Hospital	2023-12-28	16	MALE	2023-12-20
NMC065919	Mayakie MCHP	2023-07-27	6	FEMALE	2023-07-20
NMC092898	Foredugu MCHP	2023-11-15	4	FEMALE	2023-11-15
NMC132107	Rokupr CHC	2023-08-06	40	MALE	2023-07-28
NMC144004	Connaught Hospital	2023-09-10	25	MALE	

Clinical Data from Patient Record

Case ID	Visit Date	Age	Sex	History of fever	Jaundice	Date of Onset	Onset date of fever	Date of first appearance of jaundice
NMC002815	08/16/23	16	MALE	Y	Y	08/11/23	08/09/23	08/11/23
NMC048650	11/13/23	55	MALE			11/04/23		
NMC059540	12/28/23	16	MALE	Y	Y	12/20/23	12/20/23	12/23/23
NMC065919	07/27/23	6	FEMALE	Y	Y	07/20/23	07/20/23	07/24/23
NMC092898	11/15/23	4	FEMALE	Y	Y	11/15/23	11/15/23	11/15/23
NMC132107	08/06/23	40	MALE	Y	Y	07/28/23	07/28/23	07/31/23
NMC144004	09/10/23	25	MALE	Y	Y		08/29/23	09/05/23
NMC163917	03/07/23	0	FEMALE	N	N			

Identify common fields and match rows

Individual rows of each dataset must be matched and linked together in the same row.

Example:

Case ID	Health Facility	Date seen at HF/Community
NMC002815	Kambia Government Hospital	2023-08-16
NMC048650	Connaught Hospital	2023-11-13
NMC059540	Kambia Government Hospital	2023-12-28
NMC065919	Mayakie MCHP	2023-07-27
NMC092898	Foredugu MCHP	2023-11-15
NMC132107	Rokupr CHC	2023-08-06
NMC144004	Connaught Hospital	2023-09-10



Case ID	Visit Date	Age	Sex
NMC002815	08/16/23	16	MALE
NMC048650	11/13/23	55	MALE
NMC059540	12/28/23	16	MALE
NMC065919	07/27/23	6	FEMALE
NMC092898	11/15/23	4	FEMALE
NMC132107	08/06/23	40	MALE
NMC144004	09/10/23	25	MALE

Using Excel to merge datasets

1. Open the Notification dataset in Excel.
2. Create a new worksheet for the merged dataset
3. Click on the “Data” tab in the new worksheet.
4. Select “Get Data” => “From File” => “From Excel Workbook.”
5. Select the Patient Data Excel file
6. Launch Power Query Editor
7. Select “Merge Queries” and “Merge Queries as New”
8. Select the NMC Case Id column for both datasets
9. Click OK
10. Select the columns that you want from the Patient Dataset
11. Click “Close and Load” to see the merged dataset in the blank worksheet

Actions to take AFTER merging datasets

- Check data quality and address any issues.
- Check that your dataset is structured, formatted, and organized for analysis

Case Study: Surveillance

Case Study Assignment #5

In your small group, use the next 40 minutes to respond to Questions 8, 9, and 10 using Excel:

- Review your new dataset
- Get your new dataset ready to merge
- Merge your new dataset with your case notification dataset

Be prepared to share your reponses and final merged dataset.

Where Laila was



Where Laila is now

Past State

Data collection and reporting can be unorganized, uncoordinated, and delayed.

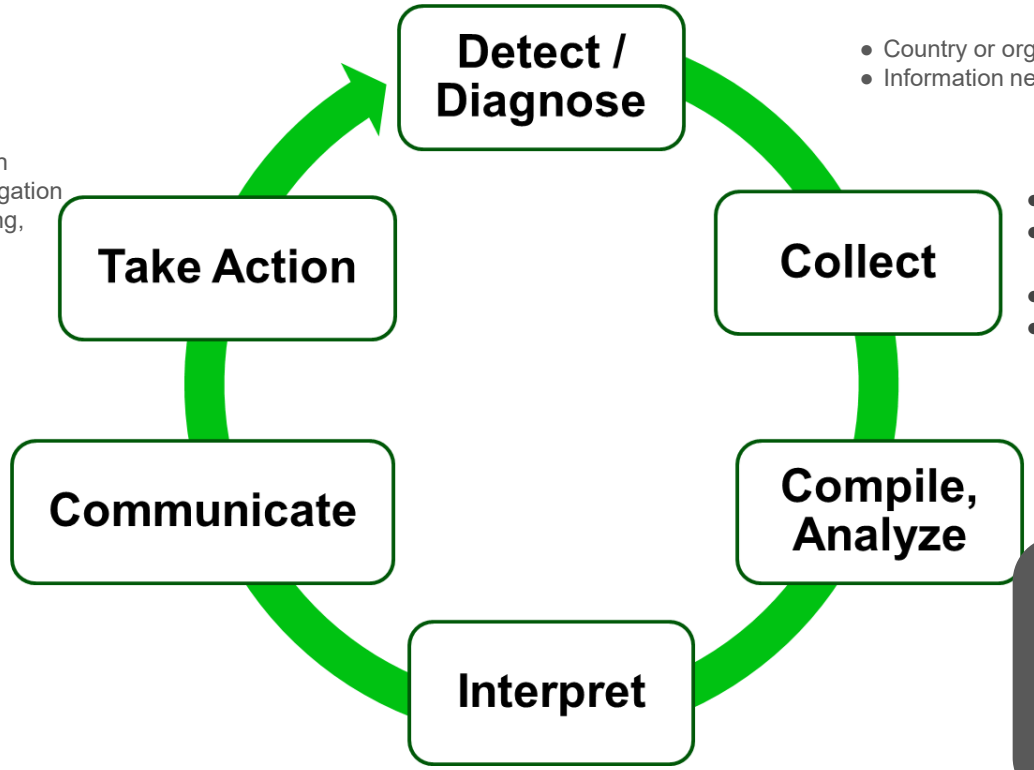
- ☑ As a result, Laila has limited access to databases and finds herself working very large datasets.
- ☑ These datasets do not always include the **quality** data she needs.
- ☑ In order to meet her information needs, she may need to locate and access additional data to address **missing** data.

Current State

Access to a dataset of quality data that

- responds to Laila's specific information needs
- can be used to summarize and analyze data

Creating Value from Public Health Surveillance Data



- Country or organizational context
- Information needs

- Data sources
- Data standards (variables, types & formats)
- Data quality
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- Describe and interpret data

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- Impact, situation awareness
- Decision on short-term action
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- Resource allocation (financing, workforce) for investigation

Where Laila is now



Laila is running out of time to prepare her surveillance summary report, which she is expected to give at this week's District Health Management Team meeting.

She needs to ***rapidly*** summarize her data in terms of person, place, and time.

She needs to interpret her findings and ***effectively*** share any recommended actions to her supervisor and other authorities.

How can informatics help Laila prepare her report and presentation?



Decisions and Actions Laila Can Take

- Select and perform the appropriate analyses
- Use the appropriate software application features or functions to run the calculations or analyses
- Use the right software application and its features to create simple and effective visualizations



Tuberculosis:

Analyzing and visualizing the data using Excel

TB surveillance goals

- Early detection of people with infectious lung disease to improve chances of clinical improvement and reduce transmission of TB
- Improve percentage of TB cases confirmed by microscope

Analyses

- Time: Graph cases and deaths monthly
- Place: Plot distribution of case households and workplaces
- Person: Count monthly cases and deaths. Analyze age and sex distribution quarterly



Tuberculosis:

Analyzing and visualizing the data using Excel

Analyses

- Time: Graph cases and deaths monthly
- Place: Plot distribution of case households and workplaces
- Person: Count monthly cases and deaths. Analyze age and sex distribution quarterly

What Excel can do

- Calculate the mode, median, mean, and range
- Create a histogram or bar chart

Use Excel to calculate measures

Measures of central location

- Mode
- Median
- Mean

Excel Statistical Functions

AVERAGE – produces the mean or average of the cells within a range of cells.

MEDIAN – produces the value closest to the midpoint of the cells within a range.

MODE -

Use Excel to calculate measures

Measure of spread

- Range

Measures of frequency

- Counts
- Ratios
- Proportions
- Rates

Excel Statistical Functions

MIN – displays the smallest value in the cell range.

MAX – displays the largest value in the cell range.

SUM – adds the values of each cell within its range.

COUNT – displays the number of cells in a range containing numbers.

Creating and using data visualizations using Excel

Data visualizations can be used to detect patterns, trends, and compare different data points.

Example:
You want to use an epi curve to display the distribution of cases by date of onset and detect an increase in the frequency of cases, suggesting an outbreak.

Using your dataset in Excel, you use the Chart feature to select your data range and generate a histogram.

Unit 2 Case Study: Surveillance

Case Study Assignment #6

In your small groups, use the next 30 minutes to respond to Questions 11-15:

- Use Excel to create tables and graphs
- Prepare to a 5 minutes presentation of your surveillance summary report

Case Study:
Surveillance

Small Group Presentations

Taking the Pulse

What are three things you learned in this unit?



What questions do you still have?

